ИИС cheat sheet ДИ

$\cos^2 x = \frac{1+\cos 2x}{2}$; $\sin^2 x = \frac{1-\cos 2x}{2}$

1. $\int x^k \arcsin x\,dx$
   $\int x^k \operatorname{arctg} x\,dx$
   $\int x^k \ln x\,dx$
   } по части с вкарване на x^k под диференциала

2. $\int_a^b f(x)\,dg(x) = f(x)\cdot g(x)\Big|_a^b - \int_a^b g(x)\,d f(x)$

3. $\int f(x)\,dg(x) = \int f(x)\cdot g'(x)\,dx$

4. $\int f(x)\,d\,kg(x) = k\int f(x)\,dg(x)$

5. $\int f(x)\,dg(x) = \int f(x)\,d\,[g(x)+C]$

6. Нека $\int f(x)\,dx = F(x)+C \Rightarrow \int_a^b f(x)\,dx = F(b)-F(a) = F(x)\Big|_a^b$

7. $\log_a b = \frac{\ln b}{\ln a}$

8 $\int x^k \sin(ax)\,dx$; $\int x^k \cos(ax)\,dx$; $\int x^k e^{ax}\,dx$ - по части: вкарваме sin, cos, e под диференциала толкова пъти, колкото е степента на x, т.е k пъти

9. $\int \cos(2x)\,dx = \frac{1}{2}\int \cos(2x)\,d\,2x = \frac{1}{2}\sin 2x + C$

10. $\int \frac{P(x)}{Q(x)}$ - рационална функция

1) ако степента на P е поне колкото степента на Q
$\deg P \geq \deg Q$, то делим P на Q с частно и остатък

2 Получената нова рационална функция разбиваме на ел. дроби.

11. $\int_a^b f(x)\,dx \overset{x=\varphi(t)}{=\!=} \int_\alpha^\beta f(\varphi(t))\,\varphi'(t)\,dt$, където
$\varphi(\alpha)=a$, $\varphi(\beta)=b$, $\varphi:[\alpha,\beta]\to[a,b]$, $t\in(\alpha,\beta) \Rightarrow \varphi(t)\in[a,b]$

12. Интеграли от рационални функции на sinx и cos x
универсална субституция
$x = 2\operatorname{arctg} t$, $t = \operatorname{tg}\frac{x}{2}$; $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$, $dx = \frac{2}{1+t^2}\,dt$

$\sin\alpha \cdot \cos\beta = \frac{\sin(\alpha+\beta)+\sin(\alpha-\beta)}{2}$

- метод на неопределените коефициенти

$$\frac{R(x)}{(x-2)\cdot(x+3)^3(x^2+2x+4)^2(x^2+2)} =$$

$$\frac{A}{x-2}+\frac{B}{x+3}+\frac{B_1}{(x+3)^2}+\frac{B_2}{(x+3)^3}+\frac{Cx+D}{x^2+2x+4}+\frac{C_1x+D_1}{(x^2+2x+4)^2}+\frac{Ex+F}{x^2+2}$$

$\ln(x^n) = n\ln x$

$I = \int x \cdot \cos x \cdot \sin x \, dx$

$\sin(2+2) = \sin 2 \cdot \cos 2 + \cos 2 \cdot \sin 2 = 2 \sin 2 \cdot \cos 2$

$\Rightarrow \int x \cdot \cos x \cdot \sin 2 \, dx = \frac{1}{2} \int x \cdot 2 \cos x \cdot \sin x \, dx = \frac{1}{2} \int x \cdot \sin 2x \, dx$

$\int x \cdot \sin 2x \cdot \cos 2x \, dx$

$\sin(x \pm 2x) = \sin x \cdot \cos 2x \pm \cos x \cdot \sin 2x$

$\sin(x+2x) = \sin(3x) = \sin x \cdot \cos 2x + \cos x \cdot \sin 2x$

$\sin(x-2x) = \sin(-x) = -\sin x = \sin x \cdot \cos 2x - \cos 2x \cdot \sin 2x$

$\Rightarrow \int x \cdot \sin 2x \cdot \cos 2x \, dx = \frac{1}{2} \int x \cdot (2 \sin x \cdot \cos 2x) \, dx = \frac{1}{2} \int x \cdot (\sin 3x - \sin x) \, dx$

$\cos(x+y) = \cos x \cdot \cos y - \sin x \cdot \sin y$

$\cos(x-y) = \cos x \cdot \cos y + \sin x \cdot \sin y$

$\sin(x+y) = \sin x \cdot \cos y + \cos x \cdot \sin y$

$\sin(x-y) = \sin x \cdot \cos y - \cos x \cdot \sin y$

$\left(\frac{u}{v}\right)' = \frac{u' \cdot v - u \cdot v'}{v^2}$

$(e^{-x})' = (-x)' \cdot e^{-x} = -e^{-x}$

$(\ln(f(x)))' = \frac{f'(x)}{f(x)}$

Производни

1. $C' = 0$

2. $(x^a)' = a \cdot x^{a-1}$

3. $(e^x)' = e^x$, $(\ln x)' = \frac{1}{x}$

$(a^x)' = a^x \ln a$

$(\log_a x)' = \frac{1}{x \ln a}$

$(x^a)' = a \cdot x^{a-1}$

$(u^v)' = u^v \left(v' \ln u + v \cdot \frac{u'}{u}\right)$

4. $(\sin x)' = \cos x$, $(\cos x)' = -\sin x$

5. $(\operatorname{tg} x)' = \frac{1}{\cos^2 x}$, $(\operatorname{ctg} x)' = -\frac{1}{\sin^2 x}$

6. $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$

7. $(\operatorname{arctg} x)' = \frac{1}{1+x^2}$, $(\operatorname{arcctg} x)' = -\frac{1}{1+x^2}$

8. $\left(\ln\left(x + \sqrt{1+x^2}\right)\right)' = \frac{1}{\sqrt{1+x^2}}$

$\left(\ln\left(x + \sqrt{x^2-1}\right)\right)' = \frac{1}{\sqrt{x^2-1}}$

$\left(\ln \sqrt{\frac{x+1}{1-x}}\right)' = \frac{1}{1-x^2}$

Производни от по-висок ред

1. $(e^x)^{(k)} = e^x$
2. $(\sin x)^{(k)} = \sin\left(x + \frac{k\pi}{2}\right)$
3. $(\cos x)^{(k)} = \cos\left(x + \frac{k\pi}{2}\right)$
4. $(x^a)^{(k)} = k!\binom{a}{k}x^{a-k} = a(a-1)\ldots(a-k+1)x^{a-k}$
5. ако $P(x)$ е полином от степен не по-висока от m, то $P^{(k)}(x) \equiv 0$ за $k > m$
6. $\left(\frac{1}{x}\right)^{(k)} = \frac{(-1)^k k!}{x^{k+1}}$
7. $(\ln x)^{(k)} = \frac{(-1)^{k-1}(k-1)!}{x^k}$

$\left[(x+a)^{-1}\right]^{(n)} = \frac{(-1)^n n!}{(x+a)^{n+1}}$

Аритметични действия

1. $(cf(x))^{(k)} = cf^{(k)}(x)$
2. $(f(x)+g(x))^{(k)} = f^{(k)}(x) + g^{(k)}(x)$
3. $(f(x)\cdot g(x))^{(k)} = \sum_{i=0}^{k}\binom{k}{i}f^{(i)}(x)\cdot g^{(k-i)}(x)$
4. $(f(ax+b))^{(k)} = a^k \cdot f^{(k)}(ax+b)$